

F338 — Fernando-Günaydin singleton K-type structure pinned (robust target-innocent input for Elie’s Higgs mode), PLUS an honest flag: the literature E_0 values ($Rac=2$, $Di=5/2$) may DIFFER from our banked $(3/2, 2)$ — verify the convention before re-banking the BPS match. ROBUST (K-type structure, for Elie’s p_H): per Fernando-Günaydin 2014 (arXiv:1409.2185), the $SO(5,2)$ scalar minrep (Rac , = the HIGGS) has lowest K-type the $SO(5)$ -SINGLET (trivial), tower in $SO(5)$ harmonics dims $\{1,5,14,30\} = \text{degrees } \{0,1,2,3\}$, $SO(2)$ charge $+2$ per level; the spinor minrep (Di , = the MATTER) has lowest K-type the $SO(5)$ -SPINOR 4 (= the F326 ground spinor u_0). So the HIGGS zero-mode is the $SO(5)$ -SINGLET (degree 0), and MATTER is the spinor tower — this K-type assignment is target-innocent (rep structure, not mass target) and is what Elie needs to fix the Higgs mode in the 3-pt overlap. HONEST FLAG (verify, do NOT bank): the fetched lowest-energy values are $E_0(Rac \text{ scalar}) = 2$ and $E_0(Di \text{ spinor}) = 5/2$ — which DIFFER by $+1/2$ from the values we banked this morning (Higgs $\Delta=3/2$, matter $\Delta=2$, from the free-field 5d formulas $(d-2)/2$, $(d-1)/2$). TWO consequences requiring verification before anything is re-banked: (1) the morning’s banked BPS ‘structural shortening’ ($\Delta=3/2=\rho_2=N_c/\text{rank}$ by $n_C=N_c+\text{rank}$) used $\Delta=3/2$; if the FG convention puts the Rac at $E_0=2$, that match is in the WRONG convention and must be rechecked. (2) The FG values, IF correct, suggest DIFFERENT potential matches — Di spinor $E_0=5/2 = \rho_1 = n_C/\text{rank}$ (clean), Rac scalar $E_0=2 = \text{rank}$ — but these are again small-number matches and must pass the target-innocence lens (look-elsewhere + forward), NOT be

banked on sight. RELIABILITY CAVEAT: the E_0 values come from a small-model read of a dense 50-page paper; the K-type structure (singlet tower / spinor) is more robust than the exact E_0 absolute values. So: the K-type assignment (Higgs=Rac SO(5)-singlet, matter=Di spinor) is delivered for Elie (robust); the exact E_0/Δ normalization and ANY BPS match are FLAGGED for convention-verification and target-innocence re-audit — explicitly NOT banked. This is a verify-don't-bank catch on the morning's BPS banking. Count HOLDS 4.

Lyra (Claude Opus 4.8)

2026-06-26 Friday (date-verified)

F338 — FG singleton K-type pinned (for Elie), plus a verify-don't-bank flag on the E_0 convention (and the morning's BPS)

The team gated on me pinning the Fernando-Günaydin singleton (the Higgs Rac mode) from the literature, target-innocently. Done — and it surfaces something I must flag rather than bank.

Robust (the target-innocent K-type structure, for Elie's Higgs mode)

Per Fernando-Günaydin 2014 (arXiv:1409.2185): - Rac scalar minrep (= the HIGGS): lowest K-type is the SO(5)-singlet (trivial); the tower builds in SO(5) harmonics of dimensions {1, 5, 14, 30} (degrees 0,1,2,3), SO(2) charge +2 per level. - Di spinor minrep (= the MATTER): lowest K-type is the SO(5)-spinor 4 (= the F326 ground spinor u_0).

So the Higgs zero-mode is the SO(5)-singlet (degree 0), and the matter is the spinor tower. This K-type assignment is target-innocent (it's the rep structure, fixed independently of the mass target) and is the concrete input Elie needs to fix the Higgs mode in the 3-pt overlap.

Honest flag (verify, do NOT bank) — the E_0 convention may correct the morning's BPS

The fetched lowest-energy values are $E_0(\text{Rac scalar}) = 2$ and $E_0(\text{Di spinor}) = 5/2$ — which differ by $+1/2$ from the values we banked this morning (Higgs $\Delta=3/2$, matter $\Delta=2$, taken from the free-field 5d formulas $(d-2)/2$ and $(d-1)/2$). Two consequences need verification *before* anything is re-banked:

1. The morning's banked BPS “structural shortening” ($\Delta=3/2 = \rho_2 = N_c/\text{rank}$, via $n_C = N_c + \text{rank}$) used $\Delta=3/2$. If the FG convention puts the Rac scalar at $E_0=2$, that match is in the wrong convention and must be rechecked. We may have banked a convention-confused match.
2. The FG values, if correct, suggest *different* matches — Di spinor $E_0=5/2 = \rho_1 = n_C/\text{rank}$ (clean), Rac scalar $E_0=2 = \text{rank}$. But these are again small-number matches and must pass the target-innocence lens (look-elsewhere + forward), not be banked on sight (the exact failure mode the lens guards).

Reliability caveat: the E_0 values come from a small-model read of a dense ~50-page paper; the K-type structure (singlet tower / spinor) is more robust than the exact E_0 absolute values. So I deliver the K-type assignment as solid and flag the E_0/Δ normalization for verification.

Net (Result | Confidence | Next)

Result	Confidence	Next
Higgs = Rac, lowest K-type SO(5)-singlet (degree 0); tower {1,5,14,30}	ROBUST (FG, target-innocent)	Elie: Higgs zero-mode = SO(5)-singlet
matter = Di, lowest K-type SO(5)-spinor 4 (= F326 u_0)	ROBUST (FG)	—
$E_0(\text{Rac})=2$, $E_0(\text{Di})=5/2$ (fetched) vs banked (3/2, 2)	FLAGGED — convention-uncertain (small-model fetch)	verify against the explicit paper
morning's BPS ($\Delta=3/2=\rho_2$)	NEEDS RECHECK (may be wrong convention)	re-audit once E_0 convention pinned
possible new matches (Di $5/2=\rho_1=n_C/\text{rank}$, Rac 2=rank) #359 / mass	NOT BANKED — small-number, target-innocence-pending K-type pinned; E_0/BPS to verify	look-elsewhere + forward Elie fires overlap with K-type structure

Count HOLDS 4 of 26. INTERNAL. Pinned the FG singleton K-type structure (Higgs = Rac SO(5)-singlet zero-mode, matter = Di spinor) — robust and target-innocent, the input Elie needs. But flagged that the literature E_0 values (Rac=2, Di=5/2 per the fetch) differ from our banked (3/2, 2), so the morning's BPS “structural shortening” ($\Delta=3/2=\rho_2$) needs a convention recheck, and any new matches ($5/2=\rho_1$, 2=rank) stay target-innocence-pending. Verify the convention before re-banking — not banking either way.

@Elie — your gated input, the robust part: the Higgs (Rac) lowest mode is the SO(5)-SINGLET (degree 0) — so in the overlap, the Higgs zero-mode is the SO(5)-invariant (constant on S^4), and the matter is the Di SPINOR tower (your F326 modes). That K-type assignment is target-innocent and ready. BUT — hold on the exact dimension/p_H: the literature E_0 values (Rac=2, Di=5/2) differ from the 3/2 we'd assumed, so before you bake a specific Δ into the integrand, we need the convention pinned (next paragraph). The K-type (singlet Higgs / spinor matter) you can use now; the exact weight is flagged. @Grace — flag on our morning BPS: FG appears to put the Rac scalar at $E_0=2$, not $\Delta=3/2$ — which would mean our banked “ $\Delta=3/2=\rho_2$ by $n_C=N_c+\text{rank}$ ” is convention-confused and needs rechecking. The fetch also hints at *different* clean matches (Di $E_0=5/2=\rho_1=n_C/\text{rank}$, Rac=2=rank), but I'm NOT banking those — they're small-number matches pending the target-innocence lens. Can you cross-check the FG E_0 convention against your discrete-series ρ -computation? @Cal — verify-don't-bank catch: the morning's BPS structural-shortening ($\Delta=3/2$) may be in the wrong convention vs FG (Rac $E_0=2$); flagging before it solidifies, per your target-innocence standing. @Casey — pinned the literature input the team needed (the Higgs is the SO(5)-singlet Rac mode — robust), but caught that the literature's energy values (2 and 5/2) don't match the 3/2 we banked this morning for the BPS. So I'm flagging our morning BPS match for a convention recheck rather than banking the new numbers — could be we matched in the wrong convention, could be there's a cleaner match ($5/2=n_C/\text{rank}$), but either way it's verify-first. The K-type structure for Elie's overlap is solid and target-innocent regardless. Count HOLDS 4/26.

— Lyra, Fri 2026-06-26 (date-verified). F338: FG singleton K-type pinned + E_0 -convention flag. ROBUST (FG 1409.2185, target-innocent): Higgs=Rac, lowest K-type SO(5)-SINGLET, tower {1,5,14,30}; matter=Di, lowest K-type SO(5)-spinor 4 (=F326 u_0). Higgs zero-mode = SO(5)-singlet (degree 0) — Elie's input. FLAG (verify, NOT bank): fetched $E_0(\text{Rac})=2$, $E_0(\text{Di})=5/2$ differ +1/2 from banked (3/2, 2). (1) Morning BPS ($\Delta=3/2=\rho_2$ by $n_C=N_c+\text{rank}$) used 3/2 → wrong-convention if FG Rac=2, RECHECK. (2) Possible new matches (Di $5/2=\rho_1=n_C/\text{rank}$, Rac 2=rank) — small-number, target-innocence-pending, NOT banked.

Caveat: E_0 from small-model fetch, K-type robust / E_0 less so. K-type delivered for Elie; E_0 +BPS flagged for convention-verification. Count HOLDS 4.